

CHEMISTRY & SUSTAINABILITY

CHEM **SUS** CHEM

ENERGY & MATERIALS

Accepted Article

Title: A Ball-Milling Enabled Reformatsky Reaction

Authors: Qun Cao, Roderick Stark, Ian Fallis, and Duncan L. Browne

This manuscript has been accepted after peer review and appears as an Accepted Article online prior to editing, proofing, and formal publication of the final Version of Record (VoR). This work is currently citable by using the Digital Object Identifier (DOI) given below. The VoR will be published online in Early View as soon as possible and may be different to this Accepted Article as a result of editing. Readers should obtain the VoR from the journal website shown below when it is published to ensure accuracy of information. The authors are responsible for the content of this Accepted Article.

To be cited as: *ChemSusChem* 10.1002/cssc.201900886

Link to VoR: <http://dx.doi.org/10.1002/cssc.201900886>

WILEY-VCH

www.chemsuschem.org

A Journal of



COMMUNICATION

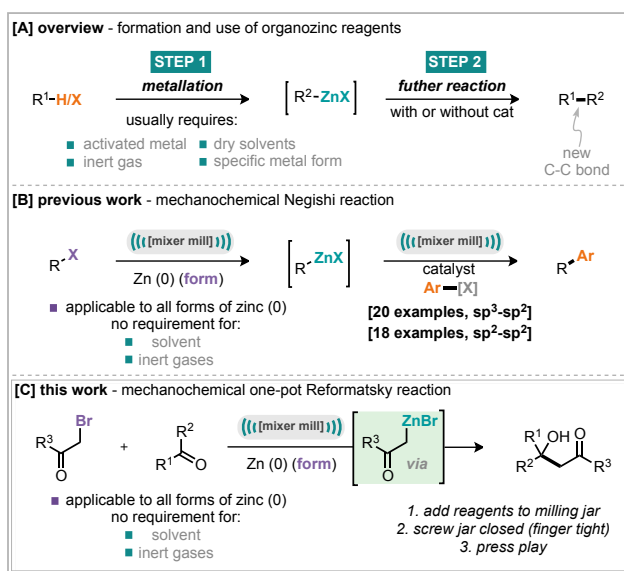
A Ball Milling Enabled Reformatsky Reaction

Qun Cao,^[a] Roderick T. Stark,^[a] Ian A. Fallis,^[a] and Duncan L. Browne*^[a]

Abstract: An operationally simple one-jar one-step mechanochemical Reformatsky reaction using *in situ* generated organozinc intermediates under neat grinding conditions has been developed. Notable features are that this reaction protocol requires no solvent, no inert gases and no pre-activation of the bulk zinc source. The developed process is demonstrated to have good substrate scope (39–82% yield), and is effective irrespective of the morphology of the zinc source.

Metal-mediated C-C bond formation is an essential tool in modern organic synthesis. Numerous reactions consisting of metal-mediated nucleophilic addition to electrophiles have been developed for the synthesis of complex organic molecules.^[1] However, the generally high basicity and/or nucleophilicity of some organometallic reagents restricts their use in late-stage modification, where sensitive functional groups may already exist in the chemical structure. Conversely, organozinc species represent a class of ‘mild’ organometallic compounds that demonstrate excellent functional group compatibility.^[2] Nevertheless, the preparation of organozinc species often requires initial access to more reactive organometallics, which are then transmetalated on to the desired organozinc reactant via metathesis with Zn (II) salts. Alternatively, activated Zn (0) can be used for the oxidative addition into carbon-halogen bonds (Scheme 1, A). In general the formation and manipulation of organometallic compounds is not particularly clean/green when considering that solvents often have to be distilled and dried prior to use; inert gases are commonly required, and, in the case of organozinc reagents, the form of the bulk metal can play an important role and chemical additives are typically required to generate the activated zinc species.^[3] Recently, we identified mechanochemistry and ball-milling as a tool for the straightforward generation of organozinc species without the requirement for carefully prepared solvents or inert gases. Under these conditions the input of mechanical energy is enough to breakdown the resilient metal oxide surface and, in the presence of the alkyl/aryl halide, generate the corresponding organozinc species (Scheme 1, B).^[4] These organozinc species may then be intercepted by opening the grinding jar, adding both catalyst and coupling partner before then running a telescoped Negishi

coupling reaction; such a process is applicable to both sp^3 - sp^3 and sp^3 - sp^2 coupling reactions. In this paper we exploit this concept to the one-jar, one-step preparation and use of organozinc species in a robust mechanochemical Reformatsky reaction protocol (Scheme 1, C). The Reformatsky reaction offers excellent potential for the formation of C-C bonds through (1) predictable C-C formation, (2) neutral reaction conditions (in comparison to obtaining the same products through aldol condensation), (3) broad functional group tolerance and (4) the ability to impart a high degree of stereocontrol.^[5]



Scheme 1. A) Formation and use of organozinc reagents; B) Previous one-jar two-step Negishi coupling; C) This work, one-jar one-step mechanochemical Reformatsky reaction.

The classical Reformatsky reaction, in which β -hydroxy esters are formed by reacting aldehydes/ketones with α -halo esters using metallic zinc, was reported in 1887.^[6] Since this seminal work a variety of latent nucleophiles and electrophiles have been studied and applied in this reaction, including, its routine use in the synthesis of complex natural products.^[7] However, in order to carry out the Reformatsky reaction, organozinc reagents must be prepared at the point of use; the majority of organozinc reagents are not commercially available.^[8] This process is can be problematic due to a layer of passivating zinc oxide which hampers the formation of organozinc species and requires removal through the treatment with chemical additives. Such additives include aqueous acid,^[9] iodine,^[10] 1,2-dibromoethane,^[11] or chlorotrimethylsilane.^[12] Highly reactive Rieke zinc is an alternative, but its preparation is non-trivial and requires the reduction of $ZnCl_2$ with alkali metals (Li, Na, K) and naphthalene.^[13] Further the air-sensitivity of this method renders the green credentials of the Rieke method poor, while derived form zinc metal are highly dependent on the physical form of zinc used, with multiple different forms of zinc commercially available

[a] School of Chemistry, Cardiff University, Main Building, Park Place, Cardiff, CF10 3AT, UK
E-mail: dlbrowne@cardiff.ac.uk

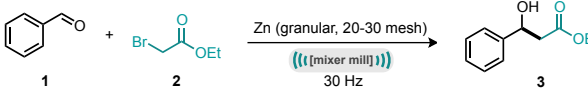
Supporting information and the ORCID identification number(s) for the author(s) of this article can be found under:
<http://XXXX>

Information about the data that underpins the results presented in this article, including how to access them, can be found in the Cardiff University data catalogue at under:
<http://XXXX> (accessed XXXX)

COMMUNICATION

(Figure S1). Thus the focus of this work is on the mechanical activation of zinc which renders the process more operationally simple and cost effective, but also delivers significant improvements of some of the green metrics typically associated with this reaction.^[14]

Mechanochemistry has been widely used among the crystal engineering and metal-organic-framework community.^[15] Recently, ball milling and other mechanochemical techniques have been explored as a method to complement the synthetic toolkit.^[16] Running reactions under mechanochemical conditions not only offers a more sustainable way to carry out solvent minimized/free reactions but can also lead to decreased reaction times, increased selectivity or different reaction outcomes when compared to results obtained from solution-based reactions.^[17] Here we describe a green protocol for the Reformatsky reaction using the ball mill mechanical activation of elemental zinc in air.



Entry	Zn [equivs]	2 [equivs]	Time [h]	Conv. [%] ^[a]	Yield [%] ^[a]
1	1.6	1.2	2	96	70
2	2	1.2	2	97	81(72%)
3	3	1.2	2	98	81
4	5	1.2	2	100	82
5	2	1.5	2	99	83
6	2	2.0	2	99	81
7	2	1.2	0.5	58	33
8	2	1.2	1	70	58
9	2	1.2	1.5	96	76
10	0	1.2	2	0	0

Reaction conditions: benzaldehyde (1 mmol), ethyl 2-bromoacetate (as specified), 20-30 mesh zinc granular (as specified), mixer mill, 30 Hz.
[a] Conversion and yield were determined by ¹H NMR using mesitylene as internal standard, isolated yield in parentheses.

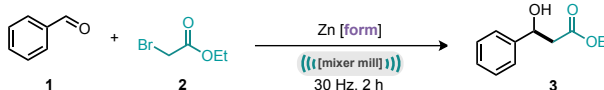
Table 1: Optimization of one-jar one-step Reformatsky reaction using ball mill.

Studies commenced by treating model substrates benzaldehyde (**1**, 1 mmol) and ethyl 2-bromoacetate (**2**, 1.2 mmol) with 1.6 equivalent of zinc (20-30 mesh zinc granular) at 30 Hz in a 10 mL grinding jar with a single ball of mass 4 grams (Table 1). After 2 hours of grinding, crude ¹H NMR analysis (mesitylene as internal standard) confirmed that there was 70% ethyl 3-hydroxy-3-phenylpropanoate (**3**) produced during the milling process (Table 1, Entry 1).

Increasing the amount of zinc to 2 equivalents resulted in 81% of the desired hydroxy-ester product **3** (Table 1, Entry 2), further increasing the amount of zinc led to no significant increase in yield (Table 1, Entries 3 and 4). Rather than increasing the equivalents of zinc, increasing the amount of ethyl 2-bromoacetate (**2**) from 1.2 to 2.0 equivalents also led to no real differences in observed yield (Table 1, Entries 5 and 6). With the optimized ratio of reagents in hand, a reaction time study assessed four individual reaction times; 0.5 h, 1 h, 1.5 h, 2 h (Table 1, Entries 2, 7-9) demonstrating that the reaction needs 2 hours to afford complete conversion. A control experiment, whereby the zinc was omitted from the reaction, returned none of the desired product.

We then applied the optimized conditions to a further 11 commercially available zinc forms (Figure S1). Although, these different zinc forms have various particle sizes, which may lead to differences in the ratio between zinc oxide layer and zinc metal, a fixed mass (2 mmol, 0.130 g) of each sample was employed. Pleasingly, it was found that in all cases the mechanochemical

Reformatsky reaction was successful irrespective of the input form of zinc (Table 2). Notably, there appears to be a general trend that the forms with a higher surface area to volume ratio perform better for the Reformatsky reaction under neat ball milling conditions, this is perhaps contrary to prediction as these metal forms should also contain a higher proportion of zinc oxide.



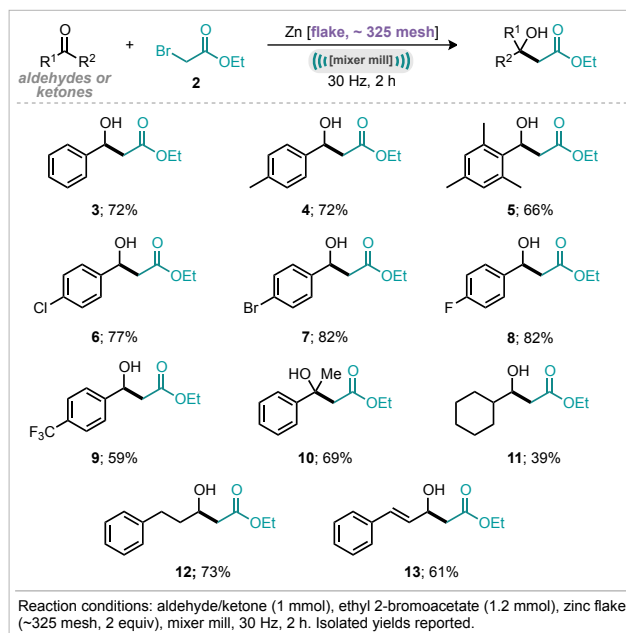
Entry	Zn [form]	Supplier	Conv. [%] ^[a]	Yield [%] ^[a]
1	zinc granular 20-30 mesh	Sigma-Aldrich	97	81
2	zinc granular 20 mesh	Sigma-Aldrich	99	72
3	zinc foil, thickness 0.25 mm 99.9%	Sigma-Aldrich	99	77
4	zinc dust < 10 mm	Sigma-Aldrich	98	80
5	zinc puriss, ACS reagent > 99.9%	Sigma-Aldrich	91	66
6	zinc shot, 10 mm (dia), 2 mm (thick), 99.99%	Alfa Aesar	87	53
7	zinc flake, ~ 325 mesh, 99.9%	Alfa Aesar	99	87
8	zinc wire (0.04 in) dia. 99.95%	Alfa Aesar	95	75
9	zinc powder 6 ~ 9 micro 97.5%	Alfa Aesar	99	83
10	zinc, metal (powder)	Fisher Scientific	98	68
11	zinc 99+% mossy	Acros	96	73
12	zinc foil, 0.38 mm	BDH chemicals	99	73

Reaction conditions: benzaldehyde (1 mmol), ethyl 2-bromoacetate (1.2 mmol), zinc form as specified (2 equiv), mixer mill, 30 Hz, 2 h.
[a] Conversion and yield were determined by ¹H NMR using mesitylene as internal standard

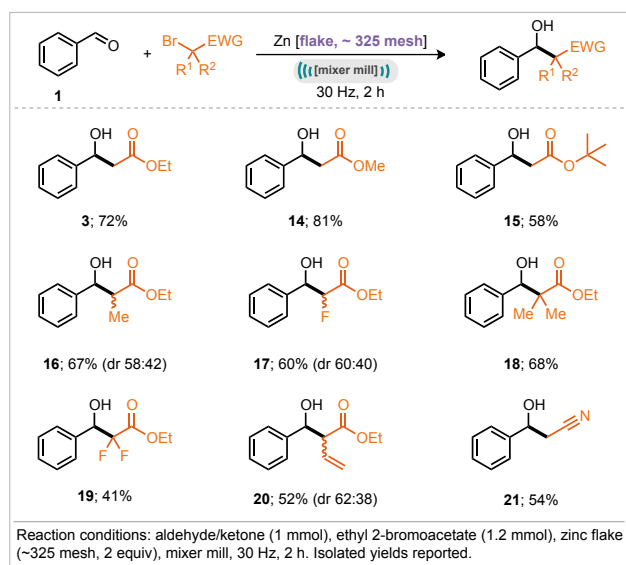
Table 2: Mechanochemical Reformatsky reaction using different zinc forms.

With the optimized conditions established, application to a small range of 11 carbonyl compounds was investigated for reactivity with ethyl 2-bromoacetate and zinc flakes (~325 mesh) under mechanical grinding (Scheme 2). It was found that the mechanochemical Reformatsky reaction demonstrates good functional group tolerance. Using highly sterically hindered **2**, **4**, 6 trimethyl benzaldehyde afforded the corresponding hydroxy-ester, **5**, in good yield (66%). Halogen containing aromatic aldehydes were also effective substrates, leading to good isolated yields of the halogen containing products (Scheme 2, **6-8**, 77-82%), hydrodehalogenation was not observed under these one-pot reaction conditions. Acetophenone was also a competent electrophile under these conditions, affording the tertiary, benzylic alcohol product, **10**, in 69% isolated yield. Aliphatic aldehydes offered mixed results with cyclohexylcarboxaldehyde giving the product in 39% yield, 2-phenylethyl aldehyde afforded 73% yield of the corresponding product and cinnamaldehyde resulted in 61% isolated yield (Scheme 2).

COMMUNICATION



Scheme 2. Scope of carbonyl electrophiles in the mechanochemical Reformatsky reaction.



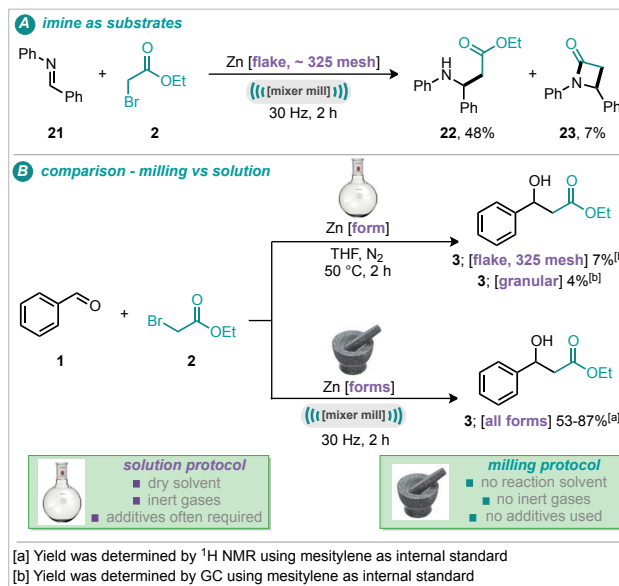
Scheme 3. Scope of latent nucleophiles in the mechanochemical Reformatsky reaction.

Eight different latent nucleophiles were also examined in this process with benzaldehyde as the model electrophile. Pleasingly, the corresponding organozinc intermediates could be generated under mechanochemical conditions (Scheme 3), both methyl and *tert*-butyl α -halo esters could be used to effectively form their β -hydroxy esters (**14** and **15**) in good to excellent yields. Reaction with ethyl 2-bromopropionate provided 67% yield of **16** as a mixture of diastereoisomers (*syn:anti* = 58:42). Notably, reaction with more stereocongested ethyl α -bromoisobutyrate maintained good yields of 68% (**17**, Scheme 3), α , α -difluoro- β -hydroxyesters **18** could also be prepared by this method. Running the ball mill

enabled Reformatsky reaction using ethyl 4-bromobut-2-enoate resulted in 52% α -substituted product **19** with moderate diastereoselectivity (*syn:anti* = 62:38). It is noteworthy that 2-bromoacetone nitrile also participated in the generation of an organozinc reagent *in situ* and formed the corresponding Reformatsky product 3-hydroxy-3-phenylpropanenitrile (**20**). Notably, in all cases explored, there was no reductive aldehyde coupling (pinacol reaction) observed.

The applicability of imines as electrophiles was also briefly explored under the developed ball milling conditions. It is known that other than aldehyde/ketone, non-classical Reformatsky electrophiles such as azomethines, nitriles, lactones, anhydrides, γ -thioalactams, amides could also be used for the Reformatsky reaction.^[18] Although not fully optimized, *N*-benzylideneaniline (**21**), did undergo a mechanochemical Reformatsky reaction (Scheme 4, A), to afford the β -amino ester, **22**, in 48% yield and a small amount of the corresponding β -lactam, **23** (7%).

The convenience of the method is well demonstrated against a solution-based reaction and that under ball milling (Scheme 4, B) using identical reagents. Running a solution based reaction in dry THF and under nitrogen gas atmosphere at 50 °C with either zinc flakes or granular zinc (no additive used) resulted in low observed yields, 7% and 4% respectively (Scheme 4, B). Whereas, the results from Table 2, show that under ball milling, without any solvent, inert gas or additive, the reaction proceeds



smoothly in two hours and all forms of zinc explored are effective for this transformation.

Scheme 4. A) Example of mechanochemical Reformatsky type reaction with imine electrophile. B) Reformatsky reaction with unactivated zinc source – milling vs solution.

In conclusion, a reliable operationally simple one-jar one-step mechanochemical Reformatsky reaction has been developed. By utilizing the organozinc generated *in situ* upon milling, this protocol avoids the requirement for access to dry solvents, inert gases and chemical additives and thus furnishes a processes where the green metrics are improved. Moreover, the protocol

COMMUNICATION

described here has developed reaction which is significantly more operationally convenient.

Acknowledgements

D.L.B is grateful to the EPSRC for a First Grant (EP/P002951/1). I.A.F. and D.L.B. are grateful to the European Union, Tenovus Cancer Charity and Cardiff University for a Knowledge Economy Skills Scholarship (KESS) to R.S. We thank the EPSRC UK National Mass Spectrometry Facility at Swansea University for mass spectrometry measurements

Conflict of interest

The authors declare no conflict of interest

Keywords: mechanochemistry • organozinc formation • ball milling • reformatsky reaction

- [1] A. Boudier, L. O. Bromm, M. Lotz and P. Knochel, *Angew. Chem. Int. Ed.*, **2000**, 39, 4414–4435
- [2] a) P. Knochel, J. Perea and P. Jones, *Tetrahedron*, **1998**, 54, 8275–8319; b) J. H. Kim, Y. O. Ko, J. Bouffard and S.-G. Lee, *Chem. Soc. Rev.*, **2015**, 44, 2489–2507.
- [3] a) P. Knochel and R. D. Singer, *Chem. Rev.*, **1993**, 93, 2117–2188; b) G. T. Achonduh, N. Hadei, C. Valente, S. Avola, C. J. O'Brien and M. G. Organ, *Chem. Commun.*, **2010**, 46, 4109–4111; c) H. N. Hunter, N. Hadei, V. Blagojevic, P. Patschinski, G. T. Achonduh, S. Avola, D. K. Bohme and M. G. Organ, *Chem. Eur. J.*, **2011**, 17, 7845–7851; d) L. C. McCann, H. N. Hunter, J. A. C. Clyburne and M. G. Organ, *Angew. Chem. Int. Ed.*, **2012**, 51, 7024–7027; e) L. C. McCann and M. G. Organ, *Angew. Chem. Int. Ed.*, **2014**, 53, 4386–4389; f) Y. H. Chen, M. Ellwart, V. Malakhov and P. Knochel, *Synthesis*, **2017**, 49, 3215–3223; g) Y. H. Chen, M. Ellwart, G. Toupalas, Y. Ebe and P. Knochel, *Angew. Chem. Int. Ed.*, **2017**, 56, 4612–4616; h) S. M. Manolikakes, M. Ellwart, C. I. Stathakis and P. Knochel, *Chem. Eur. J.*, **2014**, 20, 12289–12297.
- [4] Q. Cao, J. L. Howard, E. Wheatley and D. L. Browne, *Angew. Chem. Int. Ed.*, **2018**, 57, 11339–11343
- [5] R. Ocampo and W. R. Dolbier Jr., *Tetrahedron*, **2004**, 60, 9325–9374.
- [6] S. Reformatsky, *Ber. Dtsch. Chem. Ges.* **1887**, 20, 1210–1211.
- [7] a) A. Fürstner, *Synthesis*, **1989**, 571–590; b) S. Witkowski, Y. K. Rao, R. H. Premchandran, P. V. Halushka and J. Fried, *J. Am. Chem. Soc.*, **1992**, 114, 8464–8472; c) T. Mukaiyama, I. Shiina, H. Iwadare, M. Saitoh, T. Nishimura, N. Ohkawa, H. Sakoh, K. Nishimura, Y.-I. Tani, M. Hasegawa, K. Yamada and K. Saitoh, *Chem. Eur. J.*, **1999**, 5, 121–161; d) J. I. DeGraw, P. H. Christie, R. L. Kisliuk, Y. Gaumont and F. M. Sirotnak, *J. Med. Chem.*, **1990**, 33, 212–215.
- [8] A. Herath, V. Molteni, S. Pan and J. Loren, *Org. Lett.*, **2018**, 20, 7429–7432
- [9] M. Berton, L. Huck and J. Alcázar, *Nat Protoc.*, **2018**, 13, 324–334.
- [10] a) P. Knochel, M. C. P. Yeh, S. C. Berk and J. Talbert, *J. Org. Chem.*, **1988**, 53, 2390–2392; b) M. S. Newman, *J. Am. Chem. Soc.*, **1942**, 64, 2131–2133; c) A. K. Bose, K. Gupta and M. S. Manhas, *J. Chem. Soc., Chem. Commun.*, **1984**, 86–87; d) S. Huo, *Org. Lett.*, **2003**, 5, 423–425.
- [11] a) S. C. Berk, P. Knochel and M. C. P. Yeh, *J. Org. Chem.*, **1988**, 53, 5789–5791; b) P. Knochel and J. F. Normant, *Tetrahedron Lett.*, **1984**, 25, 1475–1478.
- [12] a) J. K. Gawroński, *Tetrahedron Lett.*, **1984**, 25, 2605–2608; b) G. Picotin and P. Miginiac, *J. Org. Chem.*, **1987**, 52, 4796–4798; c) R. Ikegami, A. Koresawa, T. Shibata and K. Takagi, *J. Org. Chem.*, **2003**, 68, 2195–2199.
- [13] a) R. D. Rieke, S. J. Uhm and P. M. Hudnall, *J. Chem. Soc., Chem. Commun.*, **1973**, 269b; b) R. D. Rieke, P. T.-J. Li, T. P. Burns and S. T. Uhm, *J. Org. Chem.*, **1981**, 46, 4323–4324; c) L. Zhu, R. M. Wehmeyer and R. D. Rieke, *J. Org. Chem.*, **1991**, 56, 1445–1453; d) A. Guijarro, D. M. Rosenberg and R. D. Rieke, *J. Am. Chem. Soc.*, **1999**, 121, 4155–4167; e) M. V. Hanson and R. D. Rieke, *J. Am. Chem. Soc.*, **1995**, 117, 10775–10776; f) R. D. Rieke, M. V. Hanson, J. D. Brown and Q. J. Niu, *J. Org. Chem.*, **1996**, 61, 2726–2730.
- [14] We are aware of a previous report of the use of a mortar and pestle to perform the Reformatsky reaction: a) K. Tanaka, S. Kishigami and F. Toda, *J. Org. Chem.*, **1991**, 56, 4333–4334; The use of a ball mill device helps to standardize this process amongst different operators, reduce the energy required from the operator and improve the safety of the reaction process, there exists a single unoptimized example of this process: b) C. Falenczyk, B. Pöllth, P. Hilgers and B. König *Synth. Commun.*, **2015**, 45, 348–354. For examples of zinc mediated modification of [60]Fullerene under ball milling conditions see: c) G.-W. Wang, Y. Murata, K. Komatsu and T. S. M. Wan, *Chem. Commun.*, **1996**, 2059–2060; d) H.-W. Liu, H. Xu, G. Shao, G.-W. Wang, *Org. Lett.* **2019**, 21, 2625–2628.
- [15] a) A. M. Belenguer, G. I. Lampronti, A. J. Cruz-Cabeza, C. A. Hunter and J. K. M. Sanders, *Chem. Sci.*, **2016**, 7, 6617–6627; b) D. Hasa, E. Miniussi and W. Jones, *Cryst. Growth Des.*, **2016**, 16, 4582–4588; c) A. V. Trask, N. Shan, W. D. S. Motherwell, W. Jones, S. Feng, R. B. H. Tan and K. J. Carpenter, *Chem. Commun.*, **2005**, 880–882; d) Y. Zhou, F. Guo, C. E. Hughes, D. L. Browne, T. R. Peskett and K. D. M. Harris, *Cryst. Growth Des.*, **2015**, 15, 2901–2907; e) D. Crawford, J. Casaban, R. Haydon, N. Giri, T. McNally and S. L. James, *Chem. Sci.*, **2015**, 6, 1645–1649; f) T. Friščić, *Chem. Soc. Rev.*, **2012**, 41, 3493–3510; g) T. Friščić, *J. Mater. Chem.*, **2010**, 20, 7599–7605; h) P. J. Beldon, L. Fábrián, R. S. Stein, A. Thirumurugan, A. K. Cheetham and T. Friščić, *Angew. Chem. Int. Ed.*, **2010**, 49, 9640–9643; i) K. Užarevič, T. C. Wang, S. Y. Moon, A. M. Fidelli, J. T. Hupp, O. K. Farha and T. Friščić, *Chem. Commun.*, **2016**, 52, 2133–2136; j) M. Klimakow, P. Klobes, A. F. Thünemann, K. Rademann and F. Emmerling, *Chem. Mater.*, **2010**, 22, 5216–5221.
- [16] a) G.-W. Wang, *Chem. Soc. Rev.*, **2013**, 42, 7668–7700; b) S. L. James, C. J. Adams, C. Bolm, D. Braga, P. Collier, T. Friščić, F. Grepioni, K. D. M. Harris, G. Hyett, W. Jones, A. Krebs, J. Mack, L. Maini, a G. Orpen, I. P. Parkin, W. C. Shearouse, J. W. Steed and D. C. Waddell, *Chem. Soc. Rev.*, **2012**, 41, 413–447; c) J.-L. Do and T. Friščić, *ACS Cent. Sci.*, **2017**, 3, 13–19; d) D. Tan and T. Friščić, *European J. Org. Chem.*, **2018**, 18–33; e) O. Eguagie, J. S. Vyle, P. F. Conlon, M. A. Gilea and Y. Liang, *Beilstein J. Org. Chem.*, **2018**, 14, 955–970; f) J. Andersen and J. Mack, *Green Chem.*, **2018**, 20, 1435–1443.
- [17] a) J. L. Howard, Q. Cao and D. L. Browne, *Chem. Sci.*, **2018**, 9, 3080–3094; b) J. G. Hernández and C. Bolm, *J. Org. Chem.*, **2017**, 82, 4007–4019.
- [18] a) S. M. Hannick and Y. Kishi, *J. Org. Chem.*, **1983**, 48, 3833–3835; b) K. Mohri, Y. Yoshida, S. Mataga, Y. Hisatsune, K. Isobe, Y. Tsuda, *Heterocycles*, **1997**, 45, 2405–2412; c) M.-X. Wang, Y. Liu, H.-Y. Gao, Y. Zhang, C.-Y. Yu, Z.-T. Huang and G. W. J. Fleet, *J. Org. Chem.*, **2003**, 68, 3281–3286; d) K. H. Schwarz, K. Kleiner, R. Ludwig and H. Schick, *J. Org. Chem.*, **1992**, 57, 4013–4015; e) J. P. Michael, C. B. de Koning and T. V. Stanbury, *Tetrahedron Lett.*, **1996**, 37, 9403–9406; f) C. Kashima, X. C. Huang, Y. Harada and A. Hosomi, *J. Org. Chem.*, **1993**, 58, 793–794; g) T. Honda, H. Wakabayashi, K. Kanai, *Chem. Pharm. Bull.*, **2002**, 50, 307–308; h) C. Baldoli, P. Del Buttero, E. Licandro, A. Papagni and T. Pilati, *Tetrahedron*, **1996**, 52, 4849–4856; i) J. C. Adrian, J. L. Barkin and L. Hassib, *Tetrahedron Lett.*, **1999**, 40, 2457–2460.

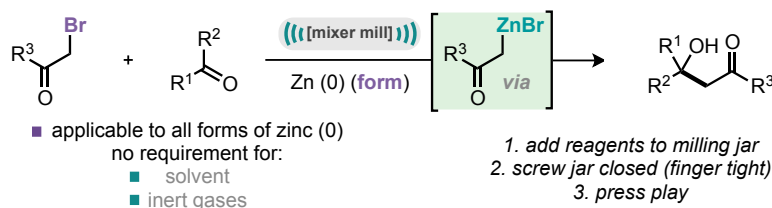
COMMUNICATION

Entry for the Table of Contents

Layout 2:

COMMUNICATION

Mechanochemical one-pot Reformatsky reaction



An operationally simple one-jar one-step mechanochemical Reformatsky reaction using *in situ* generated organozinc intermediates under neat grinding conditions has been developed. Notable features are that this reaction protocol requires no solvent, no inert gases and no pre-activation of the bulk zinc source. The developed process is demonstrated to have good substrate scope (39-82% yield), and is effective irrespective of the morphology of the zinc source.

Qun Cao, Roderick T. Stark, Ian A. Fallis, and Duncan L. Browne*

Page No. – Page No.

A Ball Milling Enabled Reformatsky Reaction